Pilot program for glider-based Passive Acoustic Monitoring of right whales in the Southeast US calving ground

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Abstract: Slocum gliders equipped with passive acoustic monitoring capabilities have been used to provide near real-time detection of right whale vocalizations for over a decade. These efforts occur in the northern foraging grounds, including the Gulf of St. Lawrence, Gulf of Maine and Southern New England. Glider-based detection has not been used in the Southeast US because right whale vocalizations are quieter in the calving ground, and glider flight operations are more complex in this shallow environment. However, right whale detection and vessel strike mitigation is especially critical in the calving ground because mother / calf pairs swim near the surface, at slow speeds, and close to shore. To address this gap in acoustic technology use, a pilot glider-based Passive Acoustic Monitoring program was started in 2023. So far, this program has consisted of a 2-week mission in January 2023, and two 4-week missions in January through March 2024. During these missions, there were 16 possible and 3 definite right whale acoustic detections. A detection on January 20, 2024 was the first definite glider-based acoustic detection in the Southeast US, eliciting near real-time detection communication via WhaleMap, Whale Alert and Robots4Whales, as well as the Early Warning System notifications program unique to the Southeast US. These pilot missions indicate that glider-based PAM may be a useful tool in supplementing aerial-based detections of right whales in the calving ground, especially providing coverage when aerial surveillance is not feasible due to weather or other logistical constraints and filling gaps in space and time.

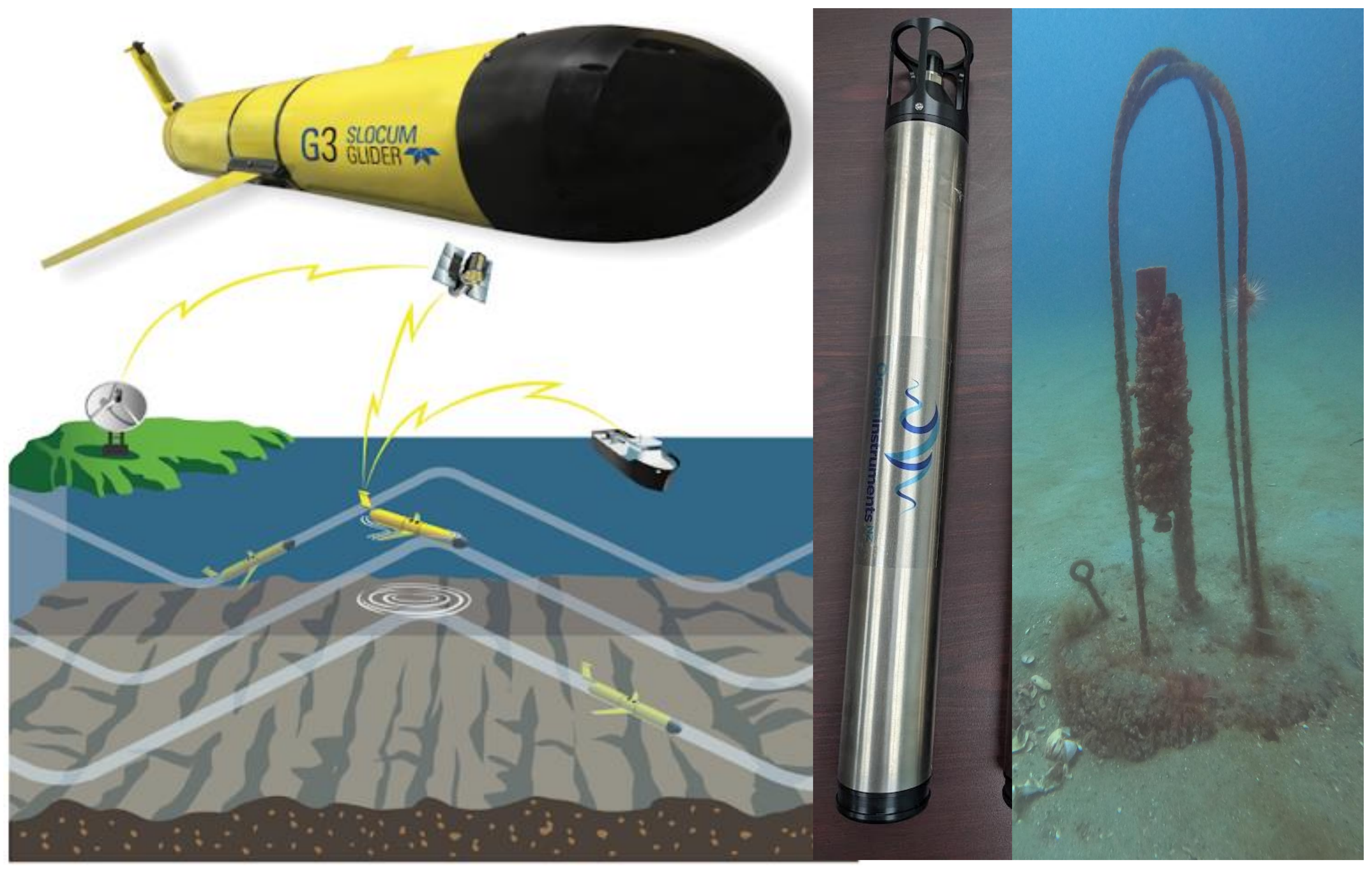

Figure 2. (above) Left: Slocum G3 Glider. Bottom: Schematic of yo-yo flight pattern and communications via cell phone and satellite Iridium link. Right: ST600 from OceanInstruments and the Sound Trap deployment configuration of our moorings upon recovery in 2023.

Methods: A Slocum G3 Glider equipped with a digital acoustic monitoring instrument (DMON2) was deployed for 3 missions off the coast of Georgia (**Fig. 1 inset; Fig. 4**). The glider was programmed to surface at 4 hour intervals to enable near real-time baleen whale vocalization detections (**Fig 2**). Three Ocean Instruments Sound Traps (ST600) were used as bottom-mounted, archival hydrophones and deployed at increasing depths (15.9m, 19.5m, 22m) and increasing distances off the coast of Savannah, Georgia (**Fig. 1, Fig. 2**). The low-frequency detection and classification system (LFDCS) was used to identify and classify calls from the real-time glider acoustic data as well as the archival hydrophone and archival glider. Detections of right whale vocalizations during the deployment period were compared between each of the three bottom-mounted hydrophones and the glider to inform optimal PAM deployment depth to detect right whales in this region.

Preliminary Results: Right whales were detected in near real-time on 3 days and possibly detected on 16 days with the glider across all 3 missions. Additional up-calls were observed on the glider and bottom-mounted archival data **(Fig. 3 & 4)**. Definite right whale detections were made on all bottom mounted hydrophones in 2023 (**Fig. 3**), with majority occurring before and after the glider was operational. Bottom mounted hydrophones in 2024 were either not recovered (shallow), corrupted (mid) or made detections (deep). Archival analysis has shown loud communities of fish that could reduce the ability to detect faint right whale calls and produce false detections in LFDCS.

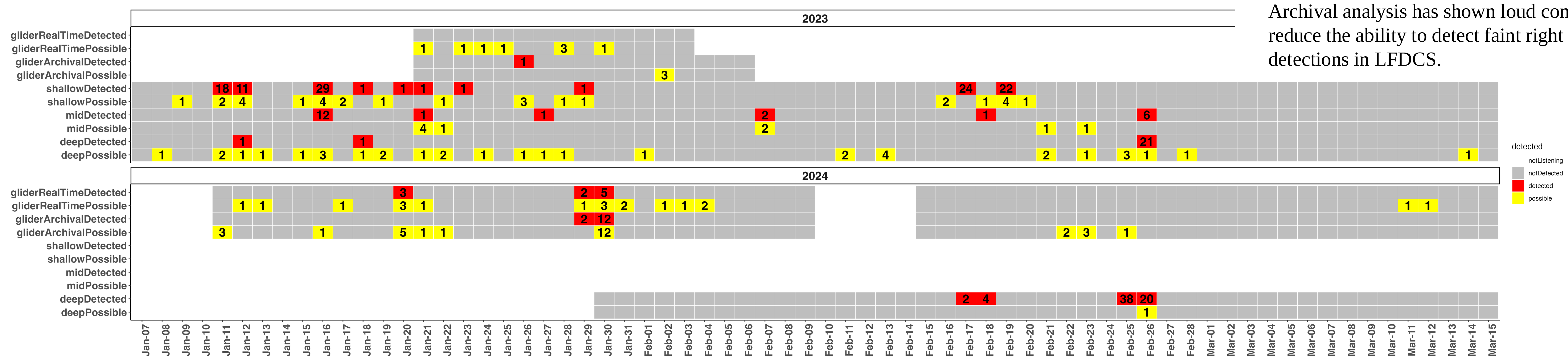

Figure 3. Preliminary analysis of right whale upcalls detected through LFDCS from the glider in near real-time, the data archived on the glider, and the bottom-mounted Sound Traps missions in 2023 (above) and 2024 (below). Definite detections shown in red and possible detections shown in yellow.

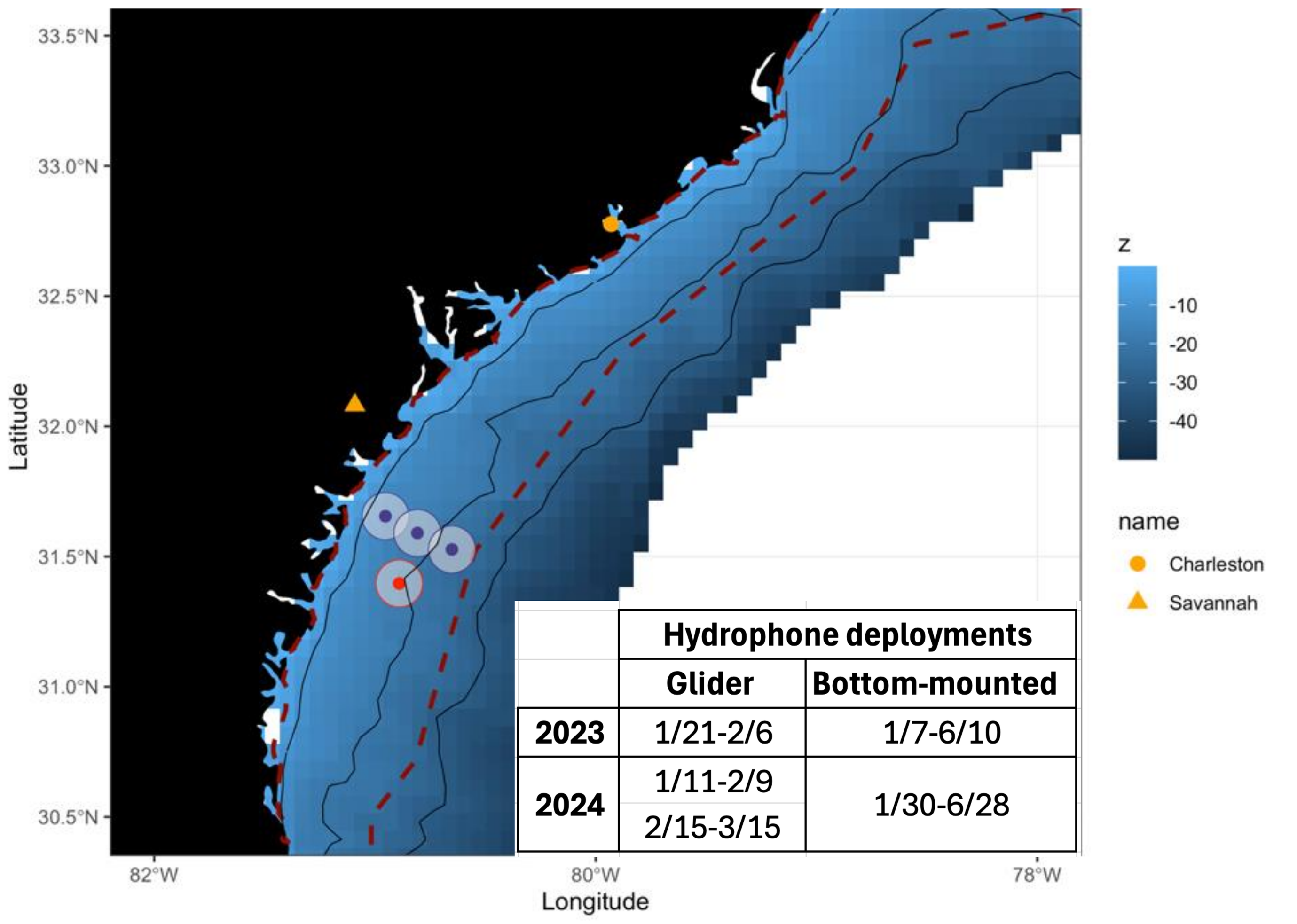

Figure 1. Locations of bottom-mounted hydrophones (purple points) and Grays Reef National Marine Sanctuary (red point). Bathymetry raster is shown in meters with contours at 10, 20 and 30m depths. Red dashed line indicates the boundaries of the North Atlantic right whale critical habitat.

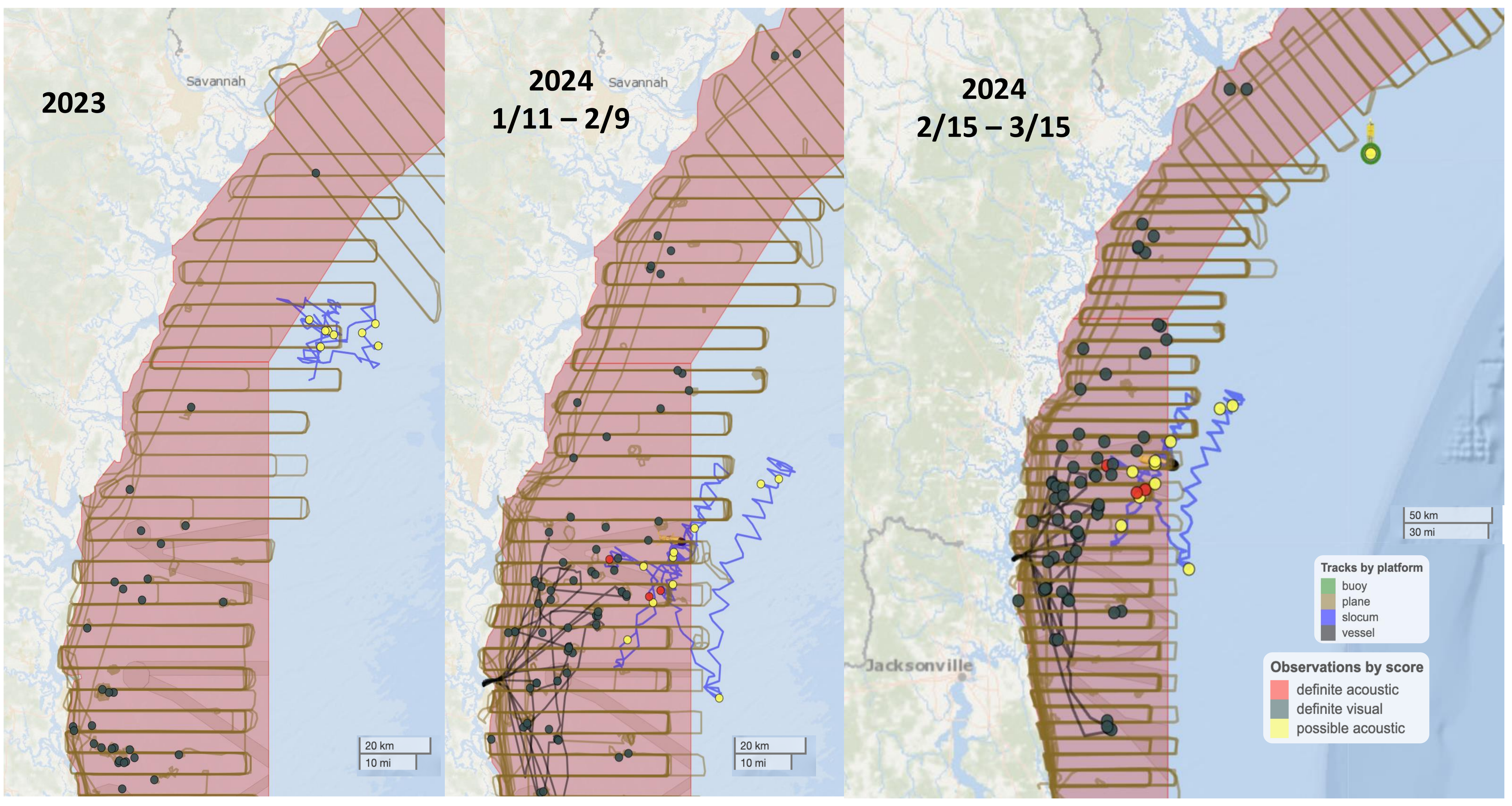

Figure 4: Glider paths are depicted in blue. The red polygon is the right whale SMA. Acoustic detections are marked in red, possible acoustic detections in yellow, and visual detections in green. Aerial and vessel survey effort is shown in light brown and dark brown respectively. Maps from WhaleMap.

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